



GuideMate

A TRUSTED NAVIGATION
PARTNER

Advance Requirement Engineering
SE 6361

Project Team



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Agenda



Purpose



As-IS Scenarios



To-Be Scenarios



Functional Requirements



Non-Functional Requirements



Traceability Matrix



Stakeholders



Venn Diagram of Specifications



Questionnaire



Prototype



Why we are Best?

Purpose

- We were given a task to develop an application helping blind people navigate indoors safely.
- Our app would rely on sensory cues like spoken directions, haptic feedback, and aural signals. Through the use of GPS, interior mapping, and object identification , our app helps blind people navigate unfamiliar places.
- The app's ultimate goal is to increase the mobility and autonomy of blind people so that they can move around in familiar and foreign areas with ease and confidence.





As-Is Problem

- John Cena, a visually impaired individual who works in a large office with multiple floors.
- He often has difficulty navigating the building, especially when trying to find meeting rooms or offices on different floors.
- He often must rely on the colleagues to guide him, which can be inconvenient and time-consuming for him and his colleagues.
- John wishes there was a better way for him to independently navigate the building and find his way to her destination without needing assistance.



As-Is Problem

- John may encounter obstacles such as furniture, office equipments or other objects left in hallways or common areas.
- These obstacles can pose safety hazards and make it difficult for John to navigate freely and independently.
- Colleagues and other building occupants might not be aware of John's visual impairment or may not understand how to assist her effectively.
- This can lead to misunderstandings and difficulties in communication.



As-Is Problem

- John may have limited access to the information about building's layout, such as floor plans or directories.
- This can make it difficult for him to plan his routes and find his way around.

To-Be Solution



- GuideMate provides a one-stop solution for John's problem.
- GuideMate gives John audio feedback asking him about his destination when tapping the phones exterior and John can confirm his destination with a voice cue.
- GuideMate can play an audio warning message while in navigation mode so that John's colleagues will be aware of his moving.
- GuideMate helps blind individuals like John become more independent and take control of their day-to-day life without depending on others.

Functional Requirements

- FR1: Securing application through fingerprint.
- FR2: Voice to provide the destination address.
- FR3: Multiple routes – Then shortest and safest.
- FR4: Guidance – Voice alerts and bell sounds or vibrations, alerts available from before 5 meter distance regarding angle of turn.
- FR5: Emergency help throughout the navigation. Using “Emergency” voice command. Call emergency contacts.
- FR6: Stores the history to navigate again.





Non-Functional Requirements

- **NFR1:** Beyond avoiding obstacles, we define “safety” in broader terms. Although utilizing many cameras and image processing techniques can enhance navigation by offering a more comprehensive view of the environment and detecting possible obstructions, it is unable to obtain data such as surface texture (wet/dry).
- **NFR2:** Even though selecting the quickest path could seem efficient, user comfort may not always follow. Users may be able to specify "comfort" according to their preferences in future updates, providing a more customized experience.
- **NFR3:** The software should be highly adaptable in order to better meet the demands of individual users. If the user feels like sharing, the app can gather information on their type of blindness to customize their navigation.
- **NFR4:** Although maintaining a clear and user-friendly interface is crucial, you should think about adding some extensibility for later updates.

Traceability Matrix

	FR1	FR2	FR3	FR4	FR5	FR6
NFR1	★			★	★	
NFR2		★	★		★	★
NFR3				★	★	
NFR4		★				

Stakeholders



- Of the people:

- Team GuideMate

- By the people:

- The people who are working to develop and deliver the system.
- Requirements engineers.
- Test engineers.
- Software architects.

- For the people:

- Blind people who need to navigate indoors.
- Caretakers who assist the blind people.
- Emergency services

Venn Diagram of specifications

Domain Properties

D1: The smartphone is pointed towards the direction the user is moving

D2: The phone is powered on

D3: The sound is enabled, and the user is able to hear audio that is played

R1: The system shall provide concise spoken instructions to avoid collision with obstacles.

Specification

C - with a CV Library as a sensor, and a speaker or earpieces as output devices

P - Program

S1: If the CV Library detects an obstacle in the path, output an audio message indicating an obstacle is present and for a new route/action.

Venn Diagram of specifications

EV (Event or Condition)	SV (System Behavior or Specification)
The smartphone is pointed towards the direction the user is moving.	The system shall use the CV Library to detect obstacles and provide audio warnings.
The phone is powered on.	The system shall remain active and functional.
The sound is enabled, and the user can hear audio.	The system shall output spoken instructions through a speaker or earpiece to guide the user.
The CV Library detects an obstacle in the path.	The system shall output an audio message indicating an obstacle is present and suggest a new route/action.

Questionnaire

- How old are you?
- On a scale from 1-5 (1 being slight visual impairment and 5 being completely blind), what is your degree of blindness?
- On a scale from 1-5 (1 being very difficult and 5 being very easy), how easily are you able to navigate your surroundings?
- Do you use a cane, guide dog, or an assistive person?
Yes, No
- On a scale from 1-5 (1 being no experience whatsoever and 5 being very experienced), how experienced are you with using smartphone apps?
- Do you have access to a stable internet connection?
Yes, No
- If you could pick one essential feature for this app what would it be?
- On a scale from 1-5 (1 being not concerned at all, 5 being very concerned), how concerned are you with security and privacy when it comes to using this app?
- Are you open to in-app purchases? Yes, No
- How would you rate the app's current navigation interface on a scale of 1 to 5, with 1 being difficult and 5 being very easy to use?
- Which of the following security features do you think are essential for the app to ensure safe indoor navigation?
 - Real-time obstacle detection, emergency alert system, user location privacy controls, secure data encryption

Questionnaire

- Which specific accessibility features would you recommend adding to improve the app's usability for blind users?
 - How well does the app adapt to different indoor environments based on your experience?
 - Which additional functionalities would you like to see incorporated into the app to enhance indoor navigation versatility?
 - Which strategies would you suggest to ensure the app remains accessible and user-friendly as it undergoes updates and improvements?
- How crucial is it for the app to seamlessly integrate with other assistive technologies for indoor navigation?
 - Which measures would you propose to enhance the app's reliability in providing accurate indoor navigation guidance?
 - How effectively does the app accommodate users with varying degrees of visual impairment?
 - What steps would you recommend to the app developers to ensure prompt identification and navigation around obstacles or hazards indoors?



Prototype

Guide Mate



Begin your journey by
saying start
navigation

Tell your Destination



Say EMERGENCY in
case of emergency

Point your camera to
start navigation



Detecting current location

Navigate to destination



No obstacle Move
forward



Obstacle Ahead
Turn left 45 degree



Destination Reached



EMERGENCY



Dial 911



Dial Tom



Dial Pooja

Thank you